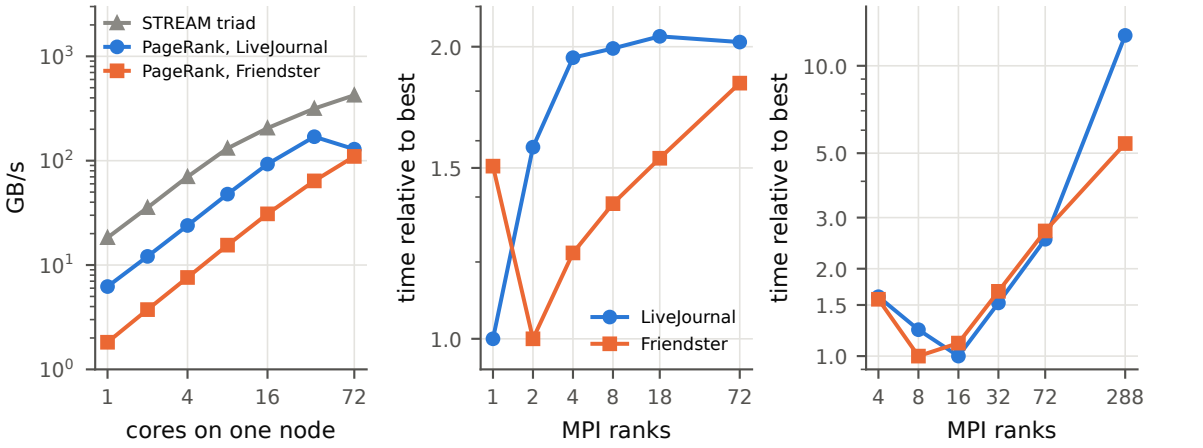
(a) Memory bandwidth
STREAM triad
PageRank, LiveJournal
PageRank, Friendster
GB/s
10^3
10^2
10^1
10^0
1
4
16
72
cores on one node
(b) 72 cores (one node)
time relative to best
2.0
1.5
1.0
LiveJournal
Friendster
1
2
4
8
18
72
MPI ranks
(c) 288 cores (four nodes)
time relative to best
10.0
5.0
3.0
2.0
1.5
1.0
4
8
16
32
72
288
MPI ranks